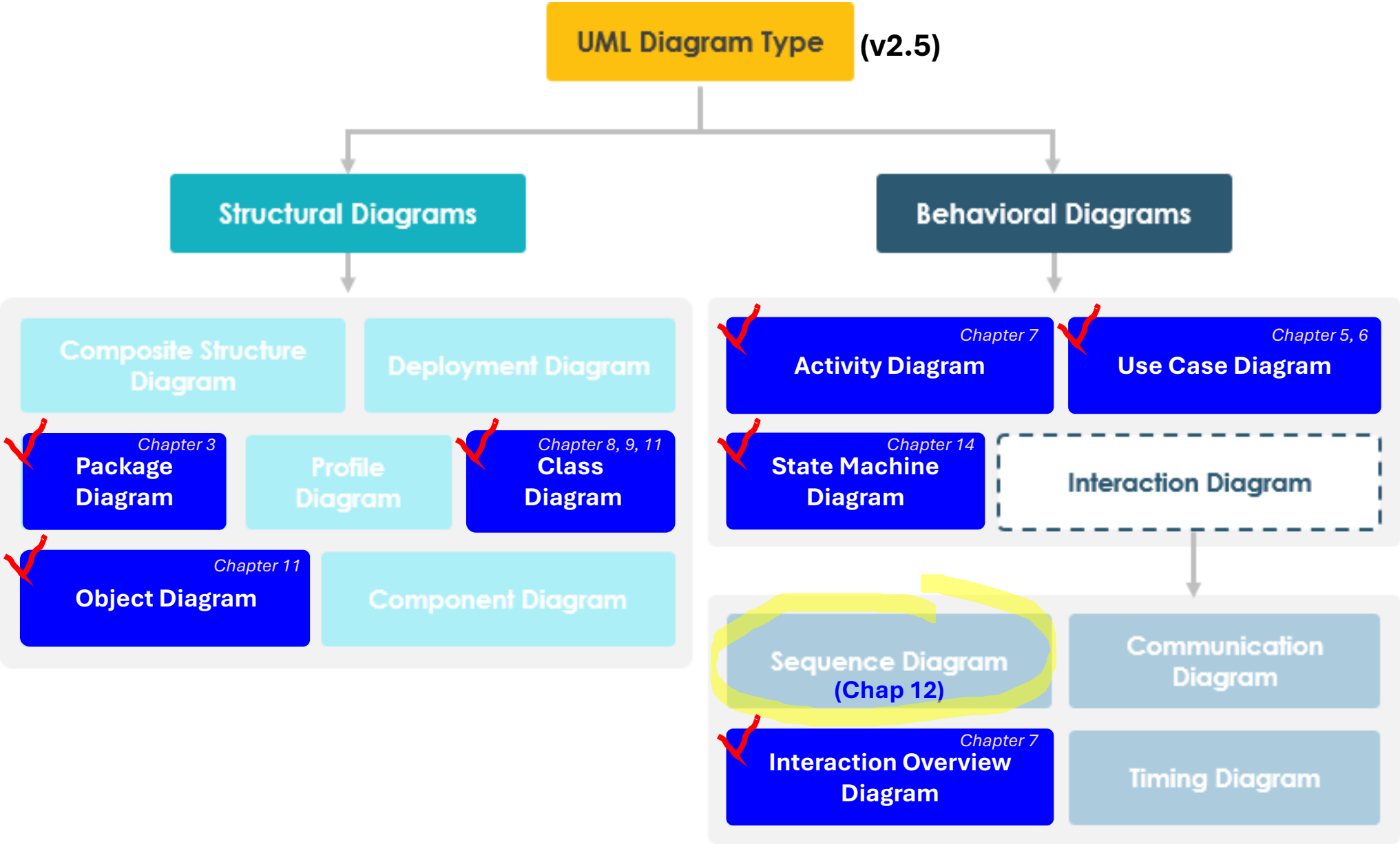
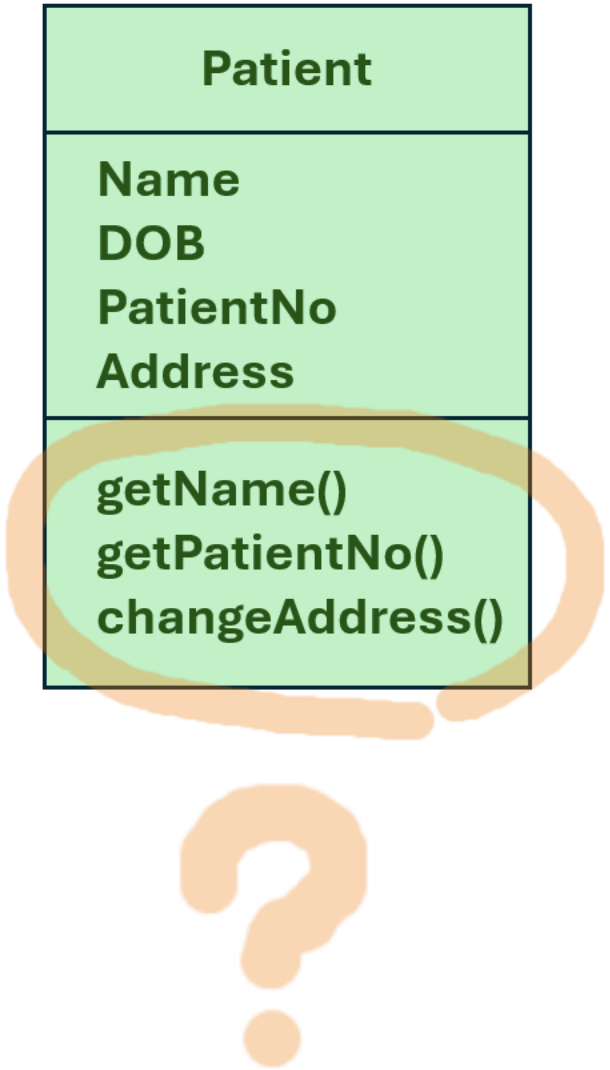


https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-interaction-overview-diagram/

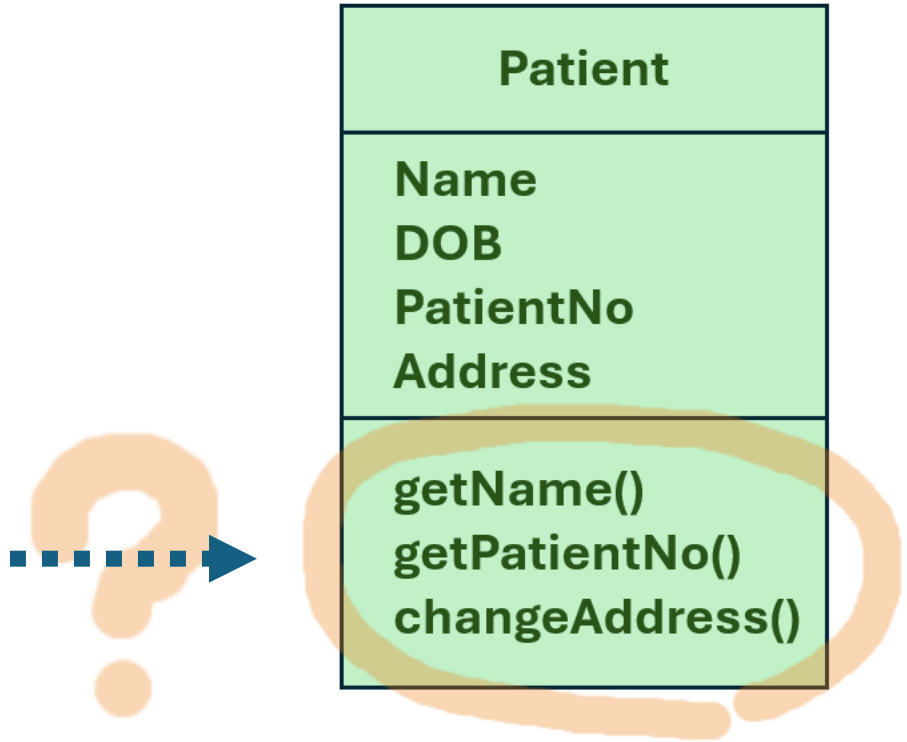
The Patient **class** and its attributes can be easily identified by analyzing use cases and/or activity diagrams.

But, how do you know **what operations or methods** need to be defined by this class?



Can any of these UML diagrams help determine the methods?

- Use cases & Use Case Diagrams *Chapter 5, 6*
- Package Diagrams *Chapter 3*
- Activity Diagrams (AD) *Chapter 7*
- Interaction Overview Diagrams (IOD) *Chapter 7*
- Class Diagrams (CD) *Chapter 8, 9, 11*



Chapter 12

Interaction Modeling with Sequence Diagrams

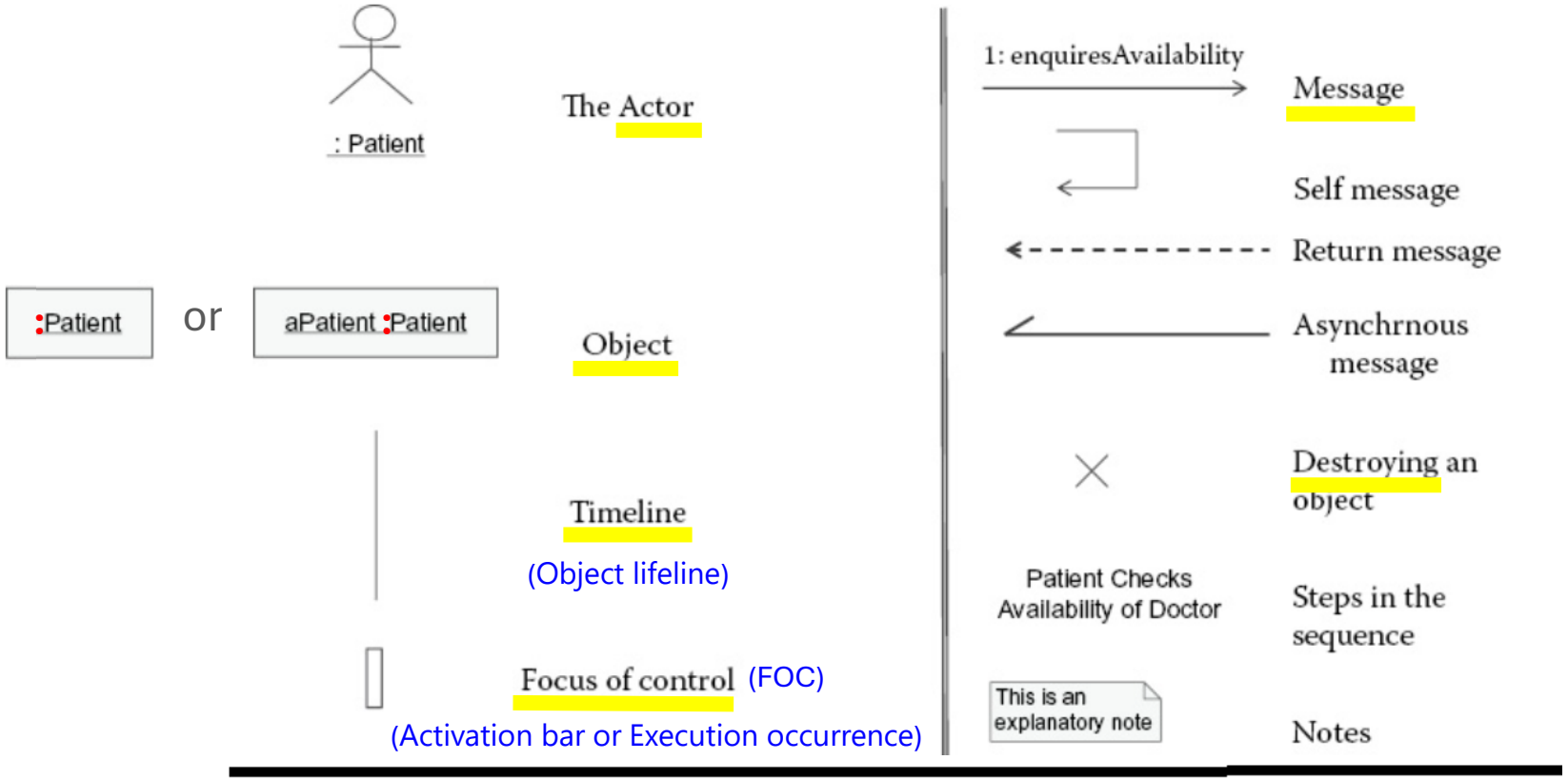


Figure 12.1 Notations of a sequence diagram.

Use Case Thumbnail: UC10-RegistersPatient

Use Case Description: This use case describes the process of registering a new patient in the hospital system. The patient must be registered and her details verified with the government health system before any of the hospital's services can be provided.

Stereotype and Package: ≪Patient≫

Preconditions: Patient must have not been already registered with the hospital.

Postconditions: Patient is registered in the hospital system.

Actors: A10-Patient; A80-Administrator; A90-GovernmentRegulatoryHealthSystem, A95-Private Health Insurance System

Use Case Relationships: Associated with actors: A10-Patient, A80-Administrator.

Basic Flow (Text):

1. A patient arrives at the hospital for some medical treatment.
2. The administrator asks the patient if she had been previously treated at this hospital.
3. Patient provides the answer (A1).
4. Administrator asks patient for her personal details such as name, address, telephone, date of birth, and emergency contact.
5. Patient provides details as requested.
6. Administrator enters details into system.
7. System verifies details (A2).
8. Administrator asks patient whether she is a public or private patient.
9. Patient provides the answer. [Public or Private][Public]
   - 9.1a Administrator asks public patient for Medicare number.
   - 9.2a Patient provides Medicare number.
   - 9.3a Administrator enters Medicare number into system.
   - 9.4a System verifies patient identity with the government health regulatory system (A3).

[Private]

   - 9.1b Administrator asks private patient for insurance details.
   - 9.2b Patient provides insurance details.
   - 9.3b Administrator enters insurance details (insurance company, patient's insurance number) into system.
   - 9.4b System verifies patient identity with private health insurance company system (A4).
10. System saves patient details.
11. System confirms patient registration.

Alternative Flow:

<A1 – If the patient has visited the hospital previously, the details are registered in the hospital's system. The administrator informs the patient of existing registration.>

<A2 – If insufficient or incorrect details have been provided, the patient is requested to provide the details again.>

<A3 – The patient cannot be verified in the government health regulatory system, so the administrator asks the patient for her Medicare card details again. If no verification is possible, the patient is conditionally registered as either a full fee–paying patient or one who will provide Medicare details later.>

<A4 – The patient cannot be verified in the private health insurance company's system, so the administrator asks the patient for the private health insurance details again. If no verification is possible, the patient is conditionally registered as either a full fee–paying patient and her details with the health insurance company are verified later.>

Exceptions: None.

Constraints: None.

User Interface Specifications: UI10-PatientRegistrationForm*

Metrics: Complex **Priority:** High **Status:** Major

Author and History: Vivek E.

Reference Material: Details of patient that are required by law are specified in the hospital's patient policy document available from the administration department. See <patientpolicy.doc>.†

Use Case
"RegistersPatient"

p.87, Chapter 5

(MOPS)

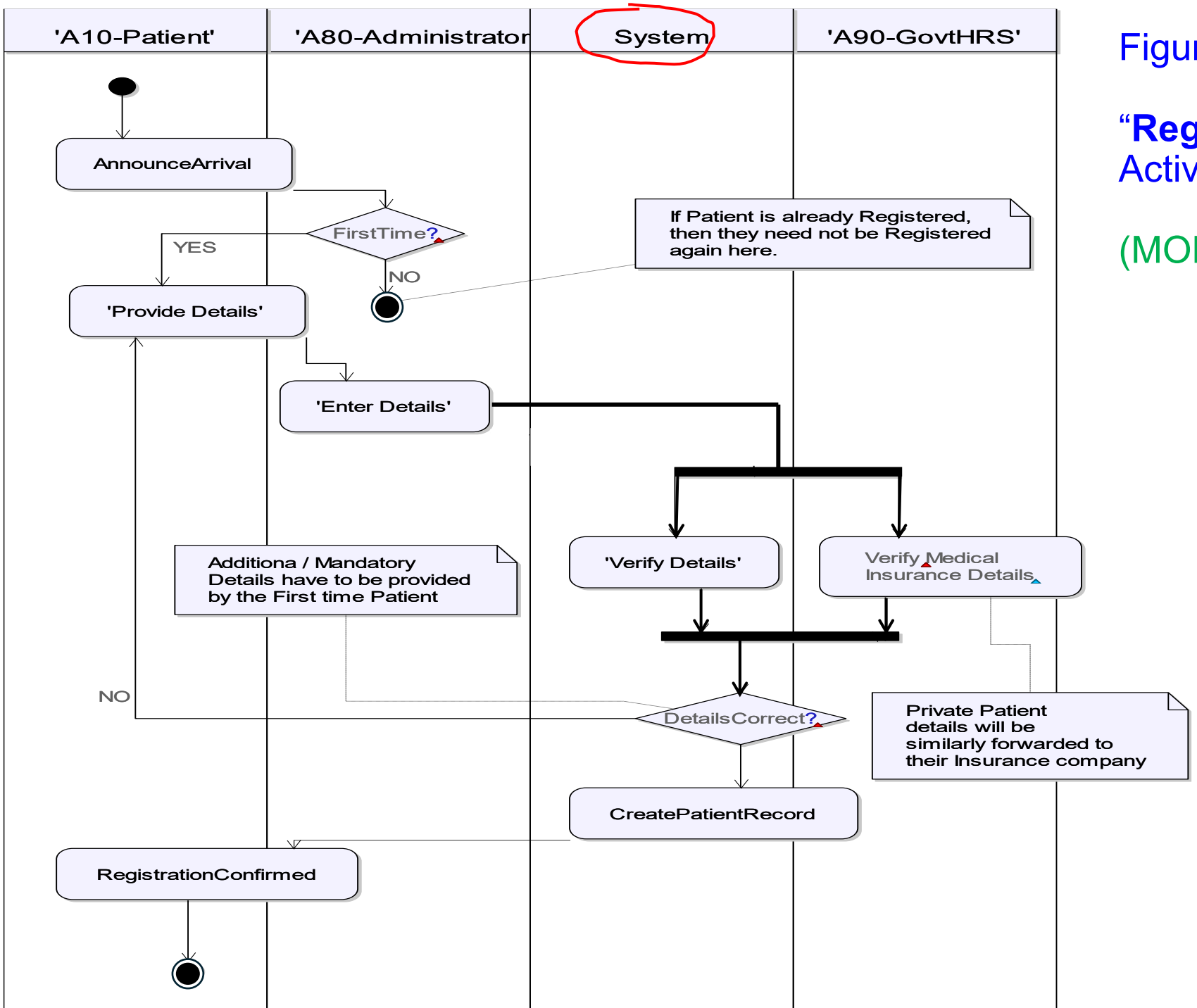


Figure 7.2

"RegistersPatient"
Activity Diagram

(MOPS)

Example of Sequence Diagram (in MOPS)

"Interactions" between actors,

between actor(s) and system,

between system and other external system(s).

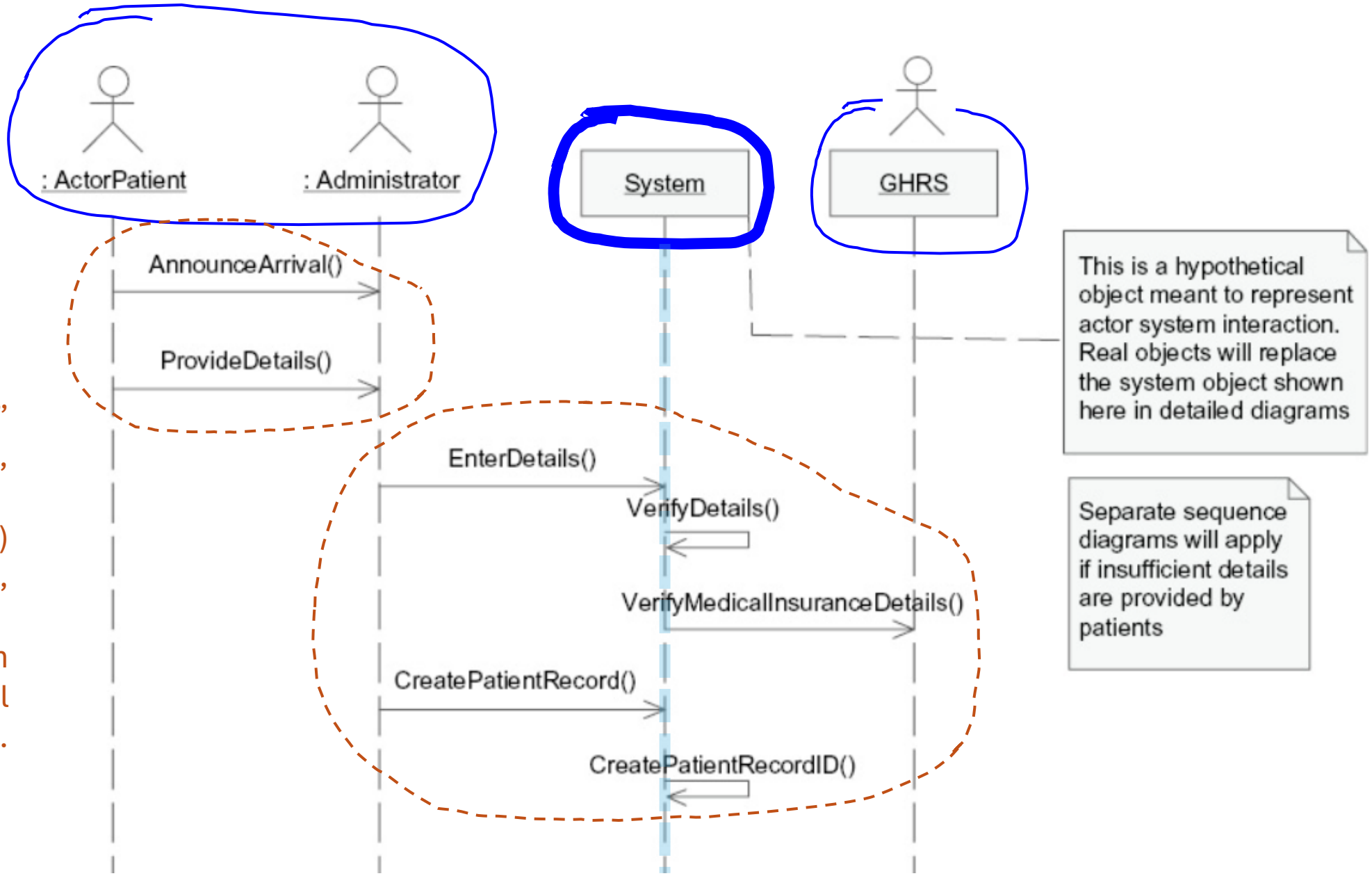
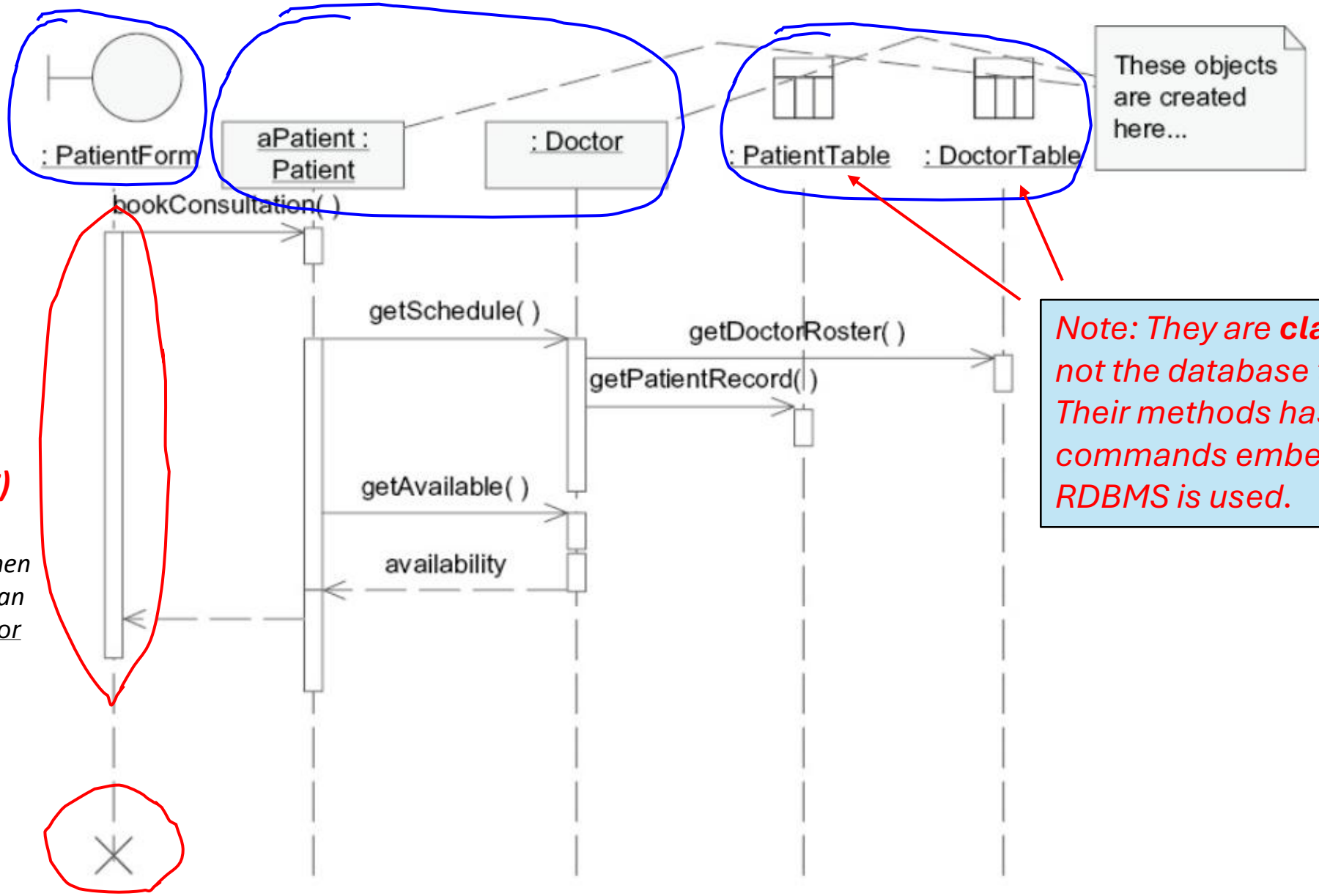


Figure 12.7 "Registering a Patient" sequence diagram.

(a use case or part of use case)

Example of advanced Sequence Diagram (in MOSS)



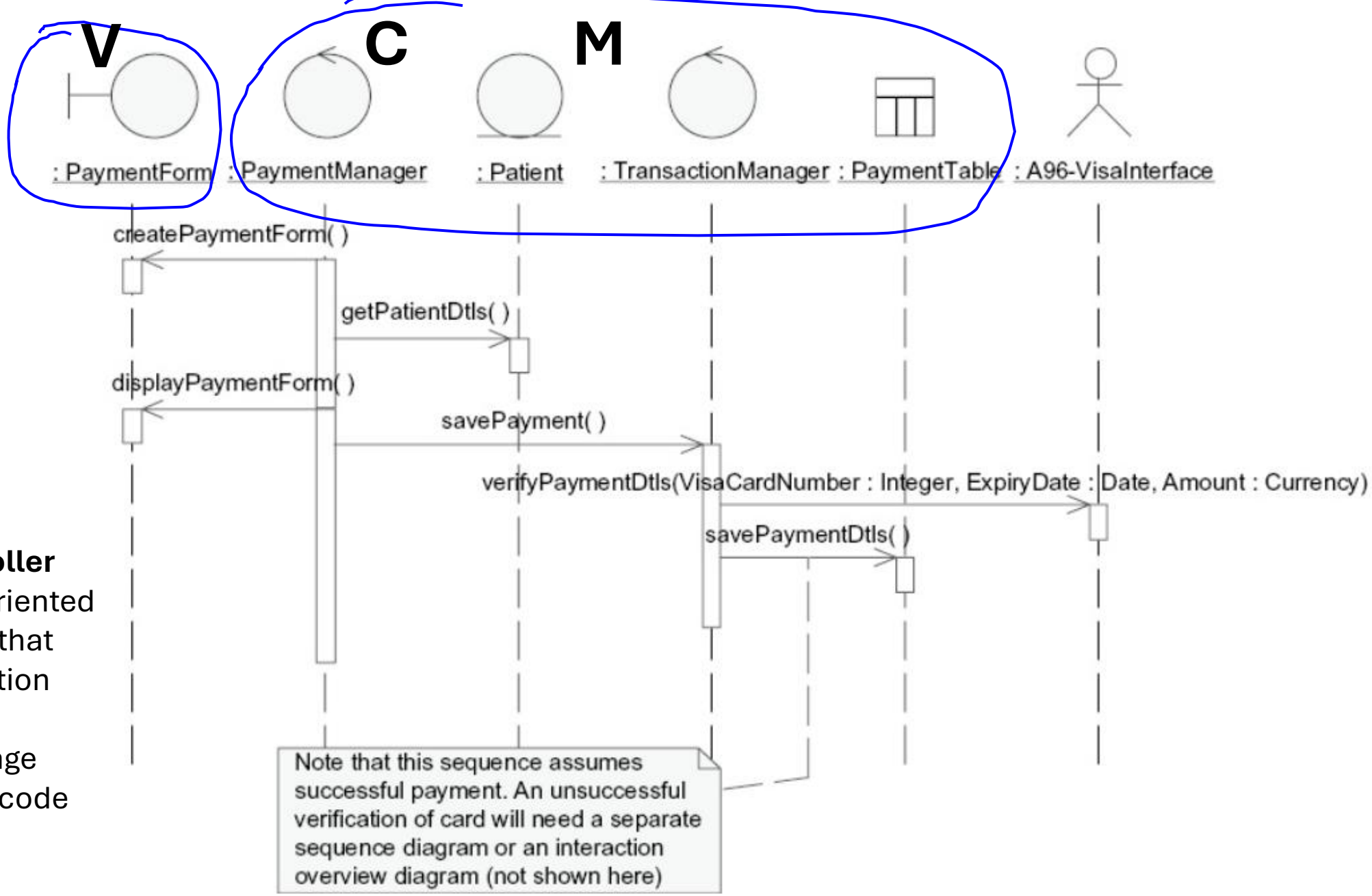
Focus of Control (FOC) *or* ***Activation bar--***
indicates the time period when an object starts performing an action <u>until its completion or return</u>

Life end of object--
indicates when the object is erased from computer memory

*Note: They are **classes**, not the database table. Their methods has SQL commands embedded, if RDBMS is used.*

Figure 12.11 A design-level sequence diagram.

Another example of advanced Sequence Diagram (in MOSS)



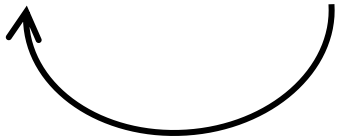
*__Model-View-Controller__ (**MVC**) is an object-oriented *architectural pattern* that separates an application into three distinct components to manage complexity, promote code reuse, and simplify maintenance.

Figure 12.15 "Paying a Bill" sequence diagram.

MVC

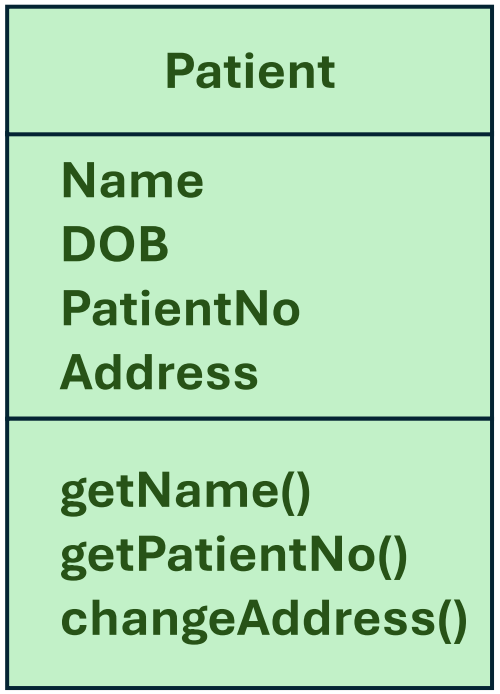
- **Model** objects - Database interaction, data validation, business rules, and state management
- **View** objects - Rendering data to the user, acting as the user interface
- **Controller** objects – “Listening” on input, *validating* user actions, *directing* the Model to update, and *selecting* a View for rendering

- V → C → M

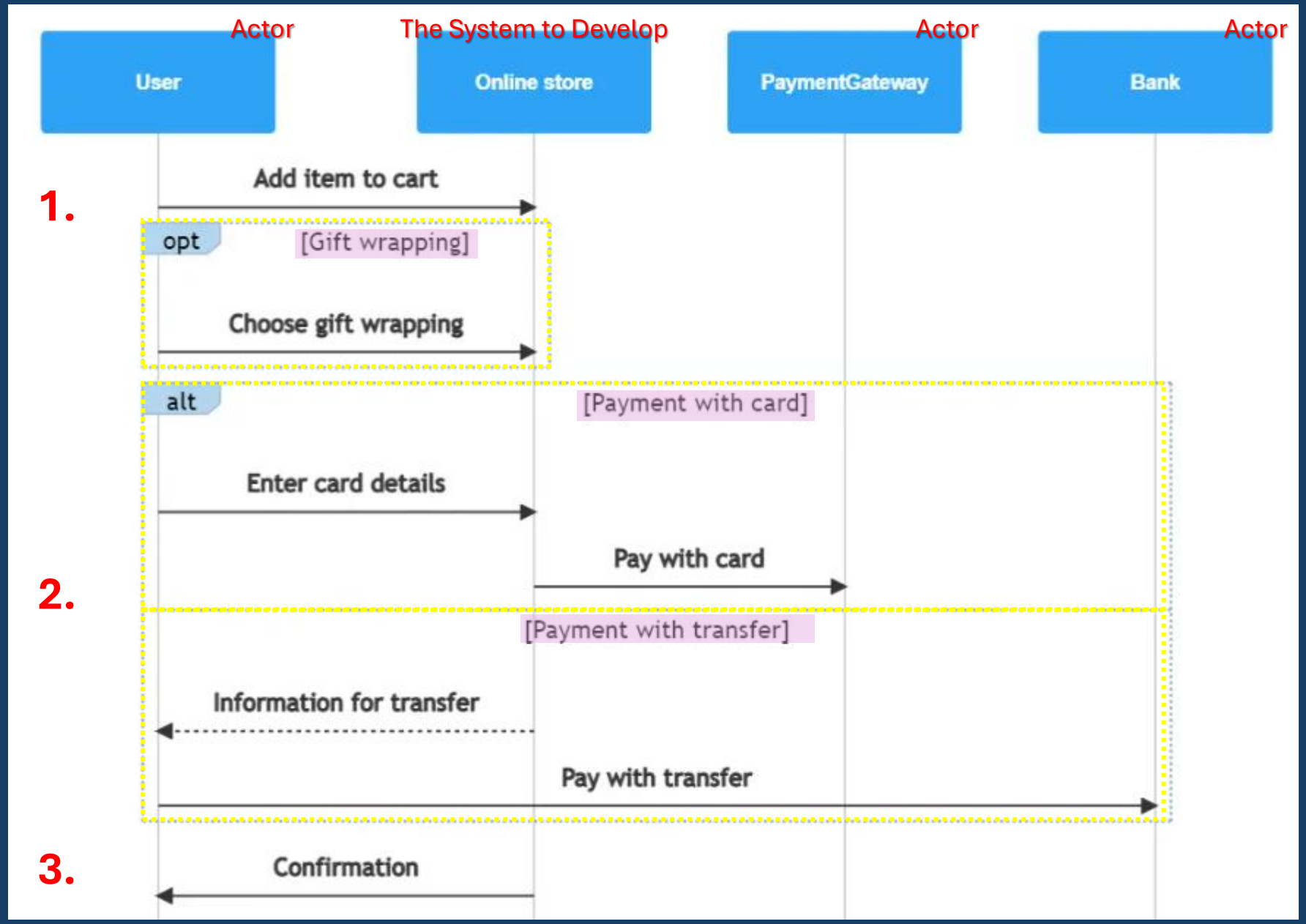


Sequence Diagrams

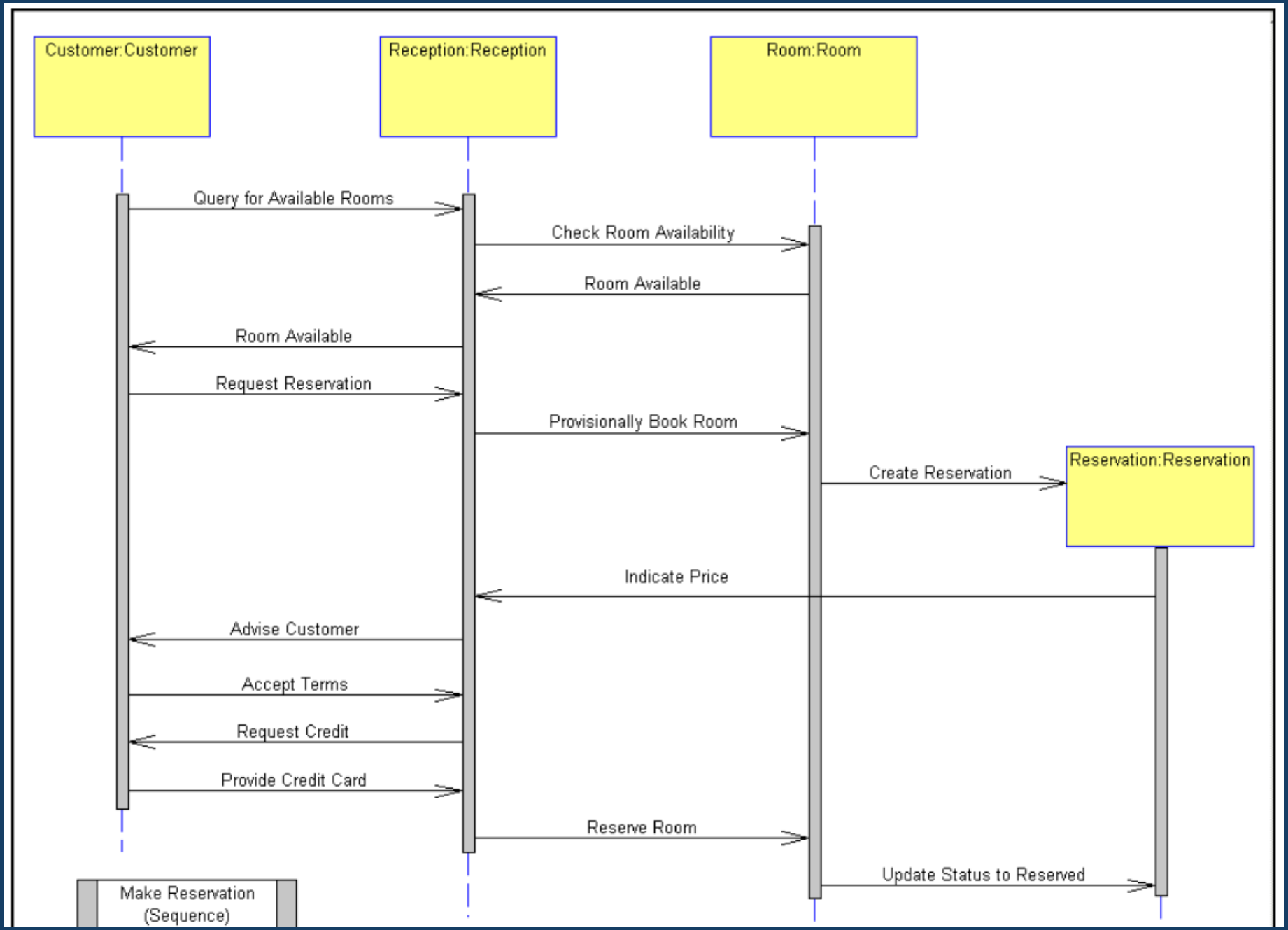
- They model and specify "interactions" between software objects in the sequential order that those interactions occur.



- Your Bellini Classes project (Project 2) may have already objects identified or designed. In this case, you need to *backtrack* to create SD for practice and documentation purpose.



https://www.gleek.io/blog/online-shopping-sequence



https://support.unicomsi.com/manuals/systemarchitect/11482/starthelp.html#page/Architecting_and_designing%2FUML.14.048.html%23